

# ROS1 fusion-hotspot benchmark report

Study: msk\_impact\_50k\_2026

## Abstract

This report analyzes ROS1 gene fusions from the msk\_impact\_50k\_2026 study, covering 122 fusion events. 72/122 fusions (59.0%) were in-frame. 107/122 fusions (87.7%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ( $p=0.214047$ , not statistically significant at  $\alpha=0.05$ ). No literature benchmark is configured for ROS1.

## Results summary

### Fusion partner frequency

Fusion-partner frequency was tabulated across 122 analyzed fusion events, identifying 30 distinct fusion partner genes. The most frequent partner was CD74, observed in 55 events.

### Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 67/72 (93.1%) of in-frame fusions retained the domain,  $p=0.214047$  (not statistically significant at  $\alpha=0.05$ ). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

### Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 68/72 (94.4%) of in-frame fusions disrupted the domain,  $p=0.0770779$  (not statistically significant at  $\alpha=0.05$ ). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.960396.

### Cutpoint detection

Cutpoint detection scanned 118 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 1693 aa (permutation-corrected  $p=0.0959041$ , not statistically significant at  $\alpha=0.05$ ). This is 25 aa from the nearest configured domain boundary at 1668 aa.

### Corroborating confidence statistics

Corroborating confidence statistics compared Domain\_retention\_flags groups "retained" ( $n=107$ ) and "not\_retained" ( $n=11$ ). For Frame\_status, retained: 67/90 (74.4%, 95% CI 64.6%-82.3%); not\_retained: 5/9 (55.6%, 95% CI 26.7%-81.1%). Welch's t-test comparing Tumor\_variant\_count between groups gave means of 68.5514 vs 19.0909,  $p=3.23984e-07$  (statistically significant at  $\alpha=0.05$ ).

### Composite evidence score

Composite evidence score produced results for this run (30 row(s) in composite\_evidence\_ranking); see the accompanying table.

## Exon retention

Exon retention produced results for this run (1 row(s) in Exon\_retention); see the accompanying table.

## Joint partner

Joint partner reported: ROS1 has no gene\_pair configured; joint-partner analysis was skipped.

# Tables

## composite\_score tables

### composite\_evidence\_ranking

| Partner_gene | Event_count | Recurrence_score    | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|---------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| CD74         | 55          | 0.45081967213114754 | 0.20043213737826426    | 0.001754963951639774    | 0.9185841308626892       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.38738259432620736 | 1    |
| SDC4         | 17          | 0.13934426229508196 | 0.20043213737826426    | 0.001754963951639774    | 0.9540766825972906       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.29926385413557793 | 2    |
| EZR          | 13          | 0.10655737704918032 | 0.20043213737826426    | 0.001754963951639774    | 0.9188947887787228       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2841505044890223  | 3    |
| COL13A1      | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | 1.0                      | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2668080894509024  | 4    |
| ZNF157       | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | 1.0                      | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2668080894509024  | 5    |
| AFG1L        | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | 0.9993891264508247       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2667164584185261  | 6    |
| SLC34A2      | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | 0.9993891264508247       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2667164584185261  | 7    |
| FN1          | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | 0.9651802076970067       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2615851206054534  | 8    |
| RBPJL        | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | 0.9651802076970067       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2615851206054534  | 9    |

| Partner_gene | Event_count | Recurrence_score    | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|---------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| NKAIN2       | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | 0.9639584605986561       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2614018585407008  | 10   |
| STX7         | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | 0.9413561392791692       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2580115103427775  | 11   |
| GOPC         | 4           | 0.03278688524590164 | 0.20043213737826426    | 0.001754963951639774    | 0.8782834453268173       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25592765543025287 | 12   |
| SLC4A4       | 2           | 0.01639344262295082 | 0.20043213737826426    | 0.001754963951639774    | 0.9022602321319487       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25460614066413734 | 13   |
| TFG          | 3           | 0.02459016393442623 | 0.20043213737826426    | 0.001754963951639774    | 0.8851557727550396       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25449948815104356 | 14   |
| IGF1R        | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | 0.9022602321319487       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2521471242706947  | 15   |
| LARS1        | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | 0.9022602321319487       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2521471242706947  | 16   |
| LRIG3        | 2           | 0.01639344262295082 | 0.20043213737826426    | 0.001754963951639774    | 0.8851557727550396       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25204047175760097 | 17   |
| NUDCD3       | 2           | 0.01639344262295082 | 0.20043213737826426    | 0.001754963951639774    | 0.8851557727550396       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25204047175760097 | 18   |
| TPM3         | 2           | 0.01639344262295082 | 0.20043213737826426    | 0.001754963951639774    | 0.8851557727550396       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25204047175760097 | 19   |

| Partner_gene | Event_count | Recurrence_score    | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|---------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| CCDC30       | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | 0.8851557727550396       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24958145536415832 | 20   |
| GOLGB1       | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | 0.8851557727550396       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24958145536415832 | 21   |
| MYH9         | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | 0.8576664630421502       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24545805890722494 | 22   |
| PRKN         | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | 0.8393402565668906       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24270912793593596 | 23   |
| CEP85L       | 2           | 0.01639344262295082 | 0.20043213737826426    | 0.001754963951639774    | 0.7486255345143555       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.23156093602149835 | 24   |
| GRIK2        | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | 0.6731826511912034       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2177854871295829  | 25   |
| NUS1         | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | 0.2229688454489921       | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.1502534162682512  | 26   |
| CERT1        | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | None                     | 0.6389004592256574         | ['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']                       | 0.13742128170694398 | 27   |
| EYS          | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | None                     | 0.6389004592256574         | ['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']                       | 0.13742128170694398 | 28   |
| NETO1        | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | None                     | 0.6389004592256574         | ['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']                       | 0.13742128170694398 | 29   |
| SLC16A10     | 1           | 0.00819672131147541 | 0.20043213737826426    | 0.001754963951639774    | 0.0                      | 0.6389004592256574         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.11680808945090237 | 30   |

## cutpoint\_detection tables

### cutpoint\_scan

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio          | p_value              | neg_log10_p_value   |
|----------|------------------------|------------------|------------------------|------------------|---------------------|----------------------|---------------------|
| 56       | 0                      | 107              | 1                      | 10               | 0.0                 | 0.09322033898305085  | 1.0304893221479003  |
| 421      | 1                      | 106              | 1                      | 10               | 0.09433962264150944 | 0.17847312762567     | 0.7484271655598992  |
| 875      | 2                      | 105              | 1                      | 10               | 0.19047619047619047 | 0.25637653793702886  | 0.5911217213946672  |
| 1430     | 3                      | 104              | 1                      | 10               | 0.28846153846153844 | 0.32750573865609567  | 0.4847810857642878  |
| 1634     | 4                      | 103              | 1                      | 10               | 0.3883495145631068  | 0.39239553580331443  | 0.4062759421356547  |
| 1688     | 5                      | 102              | 1                      | 10               | 0.49019607843137253 | 0.4515428730260006   | 0.34530100806444014 |
| 1693     | 5                      | 102              | 3                      | 8                | 0.13071895424836602 | 0.026435659744030223 | 1.577809846612513   |
| 1694     | 8                      | 99               | 3                      | 8                | 0.21548821548821548 | 0.06612721119379524  | 1.1796197925642002  |
| 1750     | 26                     | 81               | 4                      | 7                | 0.5617283950617284  | 0.4675112851720896   | 0.3302079013073733  |
| 1755     | 27                     | 80               | 4                      | 7                | 0.590625            | 0.47622392260409285  | 0.3221887920600884  |
| 1789     | 44                     | 63               | 6                      | 5                | 0.582010582010582   | 0.5242960133693759   | 0.28042344455265833 |
| 1800     | 45                     | 62               | 6                      | 5                | 0.6048387096774194  | 0.527867316527827    | 0.27747522697617977 |
| 1840     | 47                     | 60               | 6                      | 5                | 0.6527777777777778  | 0.5392135184056008   | 0.2682392283136832  |
| 1853     | 91                     | 16               | 10                     | 1                | 0.56875             | 1.0                  | -0.0                |
| 1881     | 104                    | 3                | 11                     | 0                | 0.0                 | 1.0                  | -0.0                |
| 1926     | 106                    | 1                | 11                     | 0                | 0.0                 | 1.0                  | -0.0                |

## domain\_disruption tables

### frame\_domain\_contingency\_table

|    |    |
|----|----|
| 68 | 39 |
| 4  | 7  |

### domain\_disruption\_descriptives

| Domain_key              | Domain_name                 | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|-------------------------|-----------------------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| auto_disruption_pf00041 | Fibronectin type III domain | 118                     | 11                   | 0               | 107              | 0.0                                             |

## domain\_retention tables

### frame\_domain\_contingency\_table

|    |    |
|----|----|
| 67 | 40 |
|----|----|

|   |   |
|---|---|
| 5 | 6 |
|---|---|

*domain\_retention\_descriptives*

| Domain_key | Domain_name                                  | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|------------|----------------------------------------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| kinase     | Protein tyrosine and serine/threonine kinase | 118                     | 107                  | 0               | 11               | 0.0                                             |

**exon\_retention tables**

*Exon\_retention*

| Exon | Retained_event_count | Total_event_count | Retained_fraction  |
|------|----------------------|-------------------|--------------------|
| 36   | 107                  | 122               | 0.8770491803278688 |

**frequency tables**

*Partner\_gene\_counts*

| Partner_gene | Event_count |
|--------------|-------------|
| AFG1L        | 1           |
| CCDC30       | 1           |
| CD74         | 55          |
| CEP85L       | 2           |
| CERT1        | 1           |
| COL13A1      | 1           |
| EYS          | 1           |
| EZR          | 13          |
| FN1          | 1           |
| GOLGB1       | 1           |
| GOPC         | 4           |
| GRIK2        | 1           |
| IGF1R        | 1           |
| LARS1        | 1           |
| LRIG3        | 2           |
| MYH9         | 1           |
| NETO1        | 1           |
| NKAIN2       | 1           |

| Partner_gene | Event_count |
|--------------|-------------|
| NUDCD3       | 2           |
| NUS1         | 1           |
| PRKN         | 1           |
| RBPJL        | 1           |
| SDC4         | 17          |
| SLC16A10     | 1           |
| SLC34A2      | 1           |
| SLC4A4       | 2           |
| STX7         | 1           |
| TFG          | 3           |
| TPM3         | 2           |
| ZNF157       | 1           |

## Per-event results (results.tsv)

| event_id                        | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusio<br>n_na<br>me | part<br>ner<br>_ge<br>ne | fra<br>me<br>_st<br>atu<br>s | tar<br>get<br>_ro<br>le      | break<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intron<br>ic_break<br>point | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_trun<br>cated | retain<br>ed_do<br>mains                                          | lost_<br>dom<br>ains                          | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                | source_site<br>2_effect_on<br>_frame |
|---------------------------------|---------------------------|----------------------------|---------------------|--------------------------|------------------------------|------------------------------|---------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|-------------------------------------------------------------------|-----------------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-003<br>7022-T01-I<br>M6-1 | P-0037<br>022-T0<br>1-IM6 |                            | AFG1<br>L-RO<br>S1  | AFG<br>1L                | in-fr<br>ame                 | thre<br>e_<br>_pr<br>im<br>e | 117660<br>863             | 31                          | 1694                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>tyrosin<br>e and<br>serine/<br>threoni<br>ne<br>kinase | Fibro<br>necti<br>n<br>type<br>III do<br>main |                           | LACE1 (NM_145315) -<br>ROS1 (NM_002944)<br>rearrangement: c.808-19179<br>:LACE1_c.5079+1435:ROS1<br>inv   Protein Fusion: in frame<br>{LACE1:ROS1}                    | NA                                   |
| EVT-P-004<br>1695-T01-I<br>M6-3 | P-0041<br>695-T0<br>1-IM6 |                            | CCD<br>C30-<br>ROS1 | CC<br>DC3<br>0           | out-<br>of-fr<br>ame         | thre<br>e_<br>_pr<br>im<br>e | 117643<br>359             | 35                          | 1881                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>tyrosin<br>e and<br>serine/<br>threoni<br>ne<br>kinase | Fibro<br>necti<br>n<br>type<br>III do<br>main |                           | CCDC30 (NM_001080850) -<br>ROS1 (NM_002944) fusion: t<br>(1;6)(p34.2;q22.1)(chr1:g.43<br>087864::chr6:g.117643359)  <br>Protein Fusion: out of frame<br>{CCDC30:ROS1} | NA                                   |
| EVT-P-001<br>4506-T02-I<br>M6-6 | P-0014<br>506-T0<br>2-IM6 |                            | CD74<br>-ROS<br>1   | CD7<br>4                 | unk<br>now<br>n              | thre<br>e_<br>_pr<br>im<br>e | 117650<br>623             | 32                          | 1750                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>tyrosin<br>e and<br>serine/<br>threoni<br>ne<br>kinase | Fibro<br>necti<br>n<br>type<br>III do<br>main |                           | CD74 (NM_001025159) -<br>ROS1 (NM_002944) fusion: t<br>(5;6)(q32;q22.1)(chr5:g.1497<br>82872::chr6:g.117650623)  <br>Protein Fusion: mid-exon<br>{CD74:ROS1}          | NA                                   |



| event_id                         | sample_id                 | patient_id | fusion_name       | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                | disrupted_domains | source_annotation_text                                                                                                                                                                 | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|------------|-------------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|-----------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-001<br>1025-T02-I<br>M6-7  | P-0011<br>025-T0<br>2-IM6 |            | CD74<br>-ROS<br>1 | CD7<br>4     | in-frame     | threonine_prime | 117645<br>889      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783756::chr6:g.117645889)   Protein Fusion: in frame {CD74:ROS1}                                            | NA                            |
| EVT-P-005<br>4497-T01-I<br>M6-8  | P-0054<br>497-T0<br>1-IM6 |            | CD74<br>-ROS<br>1 | CD7<br>4     | out-of-frame | threonine_prime | 117659<br>373      | 31              | 1694                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783252::chr6:g.117659373)   Protein Fusion: out of frame {CD74:ROS1}                                        | NA                            |
| EVT-P-005<br>4440-T02-I<br>M6-9  | P-0054<br>440-T0<br>2-IM6 |            | CD74<br>-ROS<br>1 | CD7<br>4     | in-frame     | threonine_prime | 117646<br>088      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782403::chr6:g.117646088)   Protein Fusion: in frame {CD74:ROS1}                                            | NA                            |
| EVT-P-005<br>4352-T01-I<br>M6-10 | P-0054<br>352-T0<br>1-IM6 |            | CD74<br>-ROS<br>1 | CD7<br>4     | out-of-frame | threonine_prime | 117649<br>394      | 33              | 1789                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783536::chr6:g.117649394)   Protein Fusion: out of frame {CD74:ROS1}                                        | NA                            |
| EVT-P-001<br>4506-T05-I<br>M7-11 | P-0014<br>506-T0<br>5-IM7 |            | CD74<br>-ROS<br>1 | CD7<br>4     | unknown      | threonine_prime | 117650<br>623      | 32              | 1750                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782872::chr6:g.117650623)   Protein Fusion: mid-exon {CD74:ROS1}                                            | NA                            |
| EVT-P-002<br>6455-T01-I<br>M6-12 | P-0026<br>455-T0<br>1-IM6 |            | CD74<br>-ROS<br>1 | CD7<br>4     | in-frame     | threonine_prime | 117646<br>566      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149783651::chr6:g.117646566)   Protein Fusion: in frame {CD74:ROS1} | NA                            |
| EVT-P-005<br>2363-T02-I<br>M7-13 | P-0052<br>363-T0<br>2-IM7 |            | CD74<br>-ROS<br>1 | CD7<br>4     | in-frame     | threonine_prime | 117647<br>256      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783700::chr6:g.117647256)   Protein Fusion: in frame {CD74:ROS1}                                            | NA                            |

| event_id                         | sample_id                 | patient_id | fusion_name | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                | disrupted_domains | source_annotation_text                                                                                                                                                                    | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|-------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|-----------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-005<br>2363-T01-I<br>M6-14 | P-0052<br>363-T0<br>1-IM6 |            | CD74-ROS1   | CD74         | in-frame     | threonine_prime | 117647256          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783692::chr6:g.117647256)   Protein Fusion: in frame {CD74:ROS1}                                               | NA                           |
| EVT-P-004<br>9640-T03-I<br>M7-15 | P-0049<br>640-T0<br>3-IM7 |            | CD74-ROS1   | CD74         | out-of-frame | threonine_prime | 117650417          | 33              | 1789                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782214::chr6:g.117650417)   Protein Fusion: out of frame {CD74:ROS1}                                           | NA                           |
| EVT-P-004<br>9640-T01-I<br>M6-16 | P-0049<br>640-T0<br>1-IM6 |            | CD74-ROS1   | CD74         | out-of-frame | threonine_prime | 117650417          | 33              | 1789                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782214::chr6:g.117650417)   Protein Fusion: out of frame {CD74:ROS1}                                           | NA                           |
| EVT-P-004<br>8671-T02-I<br>M6-17 | P-0048<br>671-T0<br>2-IM6 |            | CD74-ROS1   | CD74         | in-frame     | threonine_prime | 117647250          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783869::chr6:g.117647250)   Protein Fusion: in frame {CD74:ROS1}                                               | NA                           |
| EVT-P-001<br>1025-T01-I<br>M5-18 | P-0011<br>025-T0<br>1-IM5 |            | CD74-ROS1   | CD74         | in-frame     | threonine_prime | 117645889          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused with ROS1 exons 34-43) : t(5;6)(q32;q22.1)(chr5:g.149783755::chr6:g.117645889)   Protein fusion: in frame (CD74:ROS1) | NA                           |
| EVT-P-004<br>8671-T01-I<br>M6-19 | P-0048<br>671-T0<br>1-IM6 |            | CD74-ROS1   | CD74         | in-frame     | threonine_prime | 117647250          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783869::chr6:g.117647250)   Protein Fusion: in frame {CD74:ROS1}                                               | NA                           |
| EVT-P-004<br>7502-T01-I<br>M6-20 | P-0047<br>502-T0<br>1-IM6 |            | CD74-ROS1   | CD74         | in-frame     | threonine_prime | 117646405          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782474::chr6:g.117646405)   Protein Fusion: in frame {CD74:ROS1}                                               | NA                           |

| event_id                         | sample_id                 | patient_id | fusion_name | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                | disrupted_domains | source_annotation_text                                                                                                                                                                        | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|------------|-------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|-----------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-004<br>4465-T01-I<br>M6-21 | P-0044<br>465-T0<br>1-IM6 |            | CD74-ROS1   | CD74         | in-frame     | threonine_prime | 117646909          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783638::chr6:g.117646909)   Protein Fusion: in frame {CD74:ROS1}                                                   | NA                            |
| EVT-P-000<br>0840-T03-I<br>M6-22 | P-0000<br>840-T0<br>3-IM6 |            | CD74-ROS1   | CD74         | in-frame     | threonine_prime | 117646938          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783509::chr6:g.117646938)   Protein Fusion: in frame {CD74:ROS1}                                                   | NA                            |
| EVT-P-004<br>1118-T06-I<br>M6-23 | P-0041<br>118-T0<br>6-IM6 |            | CD74-ROS1   | CD74         | unknown      | threonine_prime | 117647425          | 33              | 1840                        | False                  | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782265::chr6:g.117647425)   Protein Fusion: mid-exon {CD74:ROS1}                                                   | NA                            |
| EVT-P-004<br>1118-T02-I<br>M6-24 | P-0041<br>118-T0<br>2-IM6 |            | CD74-ROS1   | CD74         | unknown      | threonine_prime | 117647424          | 33              | 1840                        | False                  | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782265::chr6:g.117647424)   Protein Fusion: mid-exon {CD74:ROS1}                                                   | NA                            |
| EVT-P-000<br>4747-T04-I<br>M7-25 | P-0004<br>747-T0<br>4-IM7 |            | CD74-ROS1   | CD74         | out-of-frame | threonine_prime | 117649107          | 33              | 1789                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782562::chr6:g.117649107)   Protein Fusion: out of frame {CD74:ROS1}                                               | NA                            |
| EVT-P-004<br>0869-T01-I<br>M6-26 | P-0040<br>869-T0<br>1-IM6 |            | CD74-ROS1   | CD74         | in-frame     | threonine_prime | 117647263          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783792::chr6:g.117647263)   Protein Fusion: in frame {CD74:ROS1}                                                   | NA                            |
| EVT-P-004<br>0577-T01-I<br>M6-27 | P-0040<br>577-T0<br>1-IM6 |            | CD74-ROS1   | CD74         | out-of-frame | threonine_prime | 117650159          | 33              | 1789                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused with ROS1 exons 32-43) : t(5;6)(q32;q22.1)(chr5:g.149783441::chr6:g.117650159)   Protein Fusion: out of frame {CD74:ROS1} | NA                            |

| event_id                         | sample_id                 | patient_id | fusion_name | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                | disrupted_domains | source_annotation_text                                                                                                                                                                   | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|------------|-------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|-----------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-003<br>9219-T01-I<br>M6-28 | P-0039<br>219-T0<br>1-IM6 |            | CD74-ROS1   | CD74         | in-frame     | threonine_prime | 117646339          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783535:chr6:g.117646339)   Protein Fusion: in frame {CD74:ROS1}                                               | NA                            |
| EVT-P-002<br>4572-T01-I<br>M6-29 | P-0024<br>572-T0<br>1-IM6 |            | CD74-ROS1   | CD74         | out-of-frame | threonine_prime | 117649758          | 33              | 1789                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion(CD74 exons 1-6 fused to ROS1 exons 33-43): t(5;6)(q32;q22.1)(chr5:g.149783583:chr6:g.117649758)   Protein Fusion: out of frame {CD74:ROS1} | NA                            |
| EVT-P-005<br>4667-T01-I<br>M6-30 | P-0054<br>667-T0<br>1-IM6 |            | CEP85L-ROS1 | CEP85L       | unknown      | threonine_prime | 117662314          | 30              | 1688                        | False                  | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CEP85L (NM_001178035) - ROS1 (NM_002944) fusion: c.1149-2421:CEP85L_c.5063:ROS1del   Protein Fusion: mid-exon {CEP85L:ROS1}                                                              | NA                            |
| EVT-P-006<br>3351-T02-I<br>M7-33 | P-0063<br>351-T0<br>2-IM7 |            | EZR-ROS1    | EZR          | in-frame     | threonine_prime | 117645650          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | EZR (NM_003379) - ROS1 (NM_002944) fusion: c.1091-221:EZR_c.5558-72:ROS1del   Protein Fusion: in frame {EZR:ROS1}                                                                        | NA                            |
| EVT-P-000<br>3970-T02-I<br>M5-34 | P-0003<br>970-T0<br>2-IM5 |            | EZR-ROS1    | EZR          | out-of-frame | threonine_prime | 117649698          | 33              | 1789                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | EZR (NM_003379) - ROS1 (NM_002944) fusion: (EZR exons 1 to 9 fused ROS1 exons 33 to 43): c.1090+142:EZR_c.5366_794:ROS1del   Protein fusion: out of frame (EZR-ROS1)                     | NA                            |
| EVT-P-002<br>0458-T04-I<br>M6-35 | P-0020<br>458-T0<br>4-IM6 |            | EZR-ROS1    | EZR          | in-frame     | threonine_prime | 117646033          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | EZR (NM_003379) - ROS1 (NM_002944) fusion: c.1090+208:EZR_c.5558-455:ROS1del   Protein Fusion: in frame {EZR:ROS1}                                                                       | NA                            |
| EVT-P-002<br>8081-T01-I<br>M6-36 | P-0028<br>081-T0<br>1-IM6 |            | EZR-ROS1    | EZR          | in-frame     | threonine_prime | 117647275          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | EZR (NM_003379) - ROS1 (NM_002944) fusion: c.1090+285:EZR_c.5557+112:ROS1del   Protein Fusion: in frame {EZR:ROS1}                                                                       | NA                            |

| event_id                         | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                | disrupted_domains | source_annotation_text                                                                                                                                                                        | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|-----------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-006<br>8494-T01-I<br>M7-38 | P-0068<br>494-T0<br>1-IM7 |            | FN1-<br>ROS1    | FN1          | in-frame     | threonine_prime | 117652028          | 32              | 1750                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | FN1 (NM_212482) - ROS1 (NM_002944) fusion: t(2;6)(q35;q22.1)(chr2:g.216267900::chr6:g.117652028)   Protein Fusion: in frame {FN1:ROS1}                                                        | NA                            |
| EVT-P-002<br>1147-T01-I<br>M6-40 | P-0021<br>147-T0<br>1-IM6 |            | GOPC-<br>ROS1   | GOPC         | in-frame     | threonine_prime | 117641701          | 36              | 1926                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | GOPC (NM_020399) - ROS1 (NM_002944) fusion (GOPC exons 1-4 fused with ROS1 exons 36-43) :c.651-735:GOPC_c.5778-508:ROS1del   Protein Fusion: in frame {GOPC:ROS1}                             | NA                            |
| EVT-P-001<br>5494-T01-I<br>M6-44 | P-0015<br>494-T0<br>1-IM6 |            | IGF1R-<br>ROS1  | IGF1R        | in-frame     | threonine_prime | 117646453          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | IGF1R (NM_000875) - ROS1 (NM_002944) Rearrangement : t(6;15)(q22.1,q26.3)(chr6:g.117646453::chr15:g.99477638)   Protein fusion: in frame (IGF1R-ROS1)                                         | NA                            |
| EVT-P-006<br>8483-T01-I<br>M7-45 | P-0068<br>483-T0<br>1-IM7 |            | LRIG3-<br>ROS1  | LRIG3        | in-frame     | threonine_prime | 117644742          | 35              | 1881                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | LRIG3 (NM_153377) - ROS1 (NM_002944) fusion: t(6;12)(q22.1;q14.1)(chr6:g.117644742::chr12:g.59269495)   Protein Fusion: in frame {LRIG3:ROS1}                                                 | NA                            |
| EVT-P-002<br>3007-T01-I<br>M6-46 | P-0023<br>007-T0<br>1-IM6 |            | LRIG3-<br>ROS1  | LRIG3        | in-frame     | threonine_prime | 117645256          | 35              | 1881                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | LRIG3 (NM_153377) - ROS1 (NM_002944) fusion: (LRIG3 exons 1-15 fused to ROS1 exons 35 - 43): t(6;12)(q22.1;q14.1)(chr6:g.117645256::chr12:g.59270940)   Protein Fusion: in frame {LRIG3:ROS1} | NA                            |
| EVT-P-002<br>8295-T01-I<br>M6-47 | P-0028<br>295-T0<br>1-IM6 |            | MYH9-<br>ROS1   | MYH9         | in-frame     | threonine_prime | 117641564          | 36              | 1926                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | MYH9 (NM_002473) - ROS1 (NM_002944) rearrangement: t(6;22)(q22.1;q12.3)(chr6:g.117641564::chr22:g.36691922)   Protein Fusion: in frame {MYH9:ROS1}                                            | NA                            |
| EVT-P-003<br>7864-T02-I<br>M7-49 | P-0037<br>864-T0<br>2-IM7 |            | NKAIN2-<br>ROS1 | NKAIN2       | unknown      | threonine_prime | 117662563          | 29              | 1634                        | False                  | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | NKAIN2 (NM_001040214) - ROS1 (NM_002944) fusion: c.55-122327:NKAIN2_c.4902:ROS1inv   Protein Fusion: mid-exon {NKAIN2:ROS1}                                                                   | NA                            |

| event_id                         | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role      | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                             | lost_domains                                                 | disrupted_domains | source_annotation_text                                                                                                                                                                                          | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|------------|-----------------|--------------|--------------|------------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------------------------------------|--------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-005<br>4050-T03-I<br>M6-50 | P-0054<br>050-T0<br>3-IM6 |            | NUDCD3-<br>ROS1 | NUDCD3       | out-of-frame | threep<br>_prime | 117645<br>481      | 35              | 1881                        | True                   | retained      | 1.0                      | False               | Protein<br>tyrosine<br>and<br>serine/<br>threonine<br>kinase | Fibronectin<br>type<br>III domain                            |                   | NUDCD3 (NM_015332) -<br>ROS1 (NM_002944) fusion: t<br>(6;7)(q22.1;p13)(chr6:g.1176<br>45481::chr7:g.44523353)  <br>Protein Fusion: out of frame<br>{NUDCD3:ROS1}                                                | NA                            |
| EVT-P-005<br>4050-T05-I<br>M7-51 | P-0054<br>050-T0<br>5-IM7 |            | NUDCD3-<br>ROS1 | NUDCD3       | out-of-frame | threep<br>_prime | 117645<br>481      | 35              | 1881                        | True                   | retained      | 1.0                      | False               | Protein<br>tyrosine<br>and<br>serine/<br>threonine<br>kinase | Fibronectin<br>type<br>III domain                            |                   | NUDCD3 (NM_015332) -<br>ROS1 (NM_002944) fusion: t<br>(6;7)(q22.1;p13)(chr6:g.1176<br>45481::chr7:g.44523353)  <br>Protein Fusion: out of frame<br>{NUDCD3:ROS1}                                                | NA                            |
| EVT-P-006<br>7287-T01-I<br>M7-52 | P-0067<br>287-T0<br>1-IM7 |            | PRKN-<br>ROS1   | PRKN         | unknown      | threep<br>_prime | 117674<br>184      | 26              | 1430                        | False                  | retained      | 1.0                      | False               | Protein<br>tyrosine<br>and<br>serine/<br>threonine<br>kinase | Fibronectin<br>type<br>III domain                            |                   | PARK2 (NM_004562) -<br>ROS1 (NM_002944) fusion:<br>c.171+704:PARK2_c.4290:R<br>OS1del   Protein Fusion:<br>mid-exon {PARK2:ROS1}                                                                                | NA                            |
| EVT-P-001<br>9205-T02-I<br>M6-53 | P-0019<br>205-T0<br>2-IM6 |            | RBPJL-<br>ROS1  | RBPJL        | unknown      | fivep<br>_prime  | 117654<br>098      | 31              | 1750                        | True                   | lost          | 0.0                      | False               | Fibronectin<br>type III<br>domain                            | Protein<br>tyrosine<br>and<br>serine/<br>threonine<br>kinase |                   | ROS1 (NM_002944)<br>rearrangement: t(6;20)(q22.1<br>;q13.12)(chr6:g.117654098::<br>chr20:g.43949086)  <br>Antisense Fusion                                                                                      | NA                            |
| EVT-P-002<br>1228-T01-I<br>M6-57 | P-0021<br>228-T0<br>1-IM6 |            | ROS1-<br>CD74   | CD74         | in-frame     | threep<br>_prime | 117646<br>295      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein<br>tyrosine<br>and<br>serine/<br>threonine<br>kinase | Fibronectin<br>type<br>III domain                            |                   | CD74 (NM_001025159) -<br>ROS1 (NM_002944) fusion<br>(CD74 exons 1-6 fused to<br>ROS1 exons 34-43): t(5;6)(q3<br>2;q22.1)(chr5:g.149783729::<br>chr6:g.117646295)   Protein<br>Fusion: in frame<br>{CD74:ROS1}   | NA                            |
| EVT-P-001<br>0760-T01-I<br>M5-58 | P-0010<br>760-T0<br>1-IM5 |            | ROS1-<br>CD74   | CD74         | in-frame     | threep<br>_prime | 117647<br>234      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein<br>tyrosine<br>and<br>serine/<br>threonine<br>kinase | Fibronectin<br>type<br>III domain                            |                   | CD74 (NM_001025159) -<br>ROS1 (NM_002944) fusion<br>(CD74 exons 1-6 fused to<br>ROS1 exons 34-43): t(5;6)(5<br>q32;q22.1)(chr5:g.1497837<br>63::chr6:g.117647234 )  <br>Protein fusion: in frame<br>(CD74-ROS1) | NA                            |

| event_id                 | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role  | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                                 | disrupted_domains | source_annotation_text                                                                                                                                                                     | source_site_2_effect_on_frame |
|--------------------------|-------------------|------------|-------------|--------------|--------------|--------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|----------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0002707-T03-IM5-59 | P-0002707-T03-IM5 |            | ROS1-CD74   | CD74         | in-frame     | threep_prime | 117646946          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD274 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.49783748::chr6:g.117646946)   Protein fusion: in frame (CD74-ROS1)     | NA                            |
| EVT-P-0015215-T01-IM6-60 | P-0015215-T01-IM6 |            | ROS1-CD74   | CD74         | out-of-frame | threep_prime | 117650370          | 33              | 1789                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 32-43): t(5;6)(q32;q22.1)(chr5:g.149783057::chr6:g.117650370)   Protein fusion: out of frame (CD74-ROS1) | NA                            |
| EVT-P-0002877-T01-IM3-61 | P-0002877-T01-IM3 |            | ROS1-CD74   | CD74         | in-frame     | threep_prime | 117645870          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | NA   Protein fusion: in frame (CD74-ROS1)                                                                                                                                                  | NA                            |
| EVT-P-0012525-T01-IM5-62 | P-0012525-T01-IM5 |            | ROS1-CD74   | CD74         | in-frame     | threep_prime | 117647385          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | CD74 (NM_001025159) - ROS1(NM_002944) fusion (CD74 exons 1-6 with ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149783203::chr6:g.117647385)   Protein fusion: in frame (CD74-ROS1)          | NA                            |
| EVT-P-0002912-T01-IM3-63 | P-0002912-T01-IM3 |            | ROS1-CD74   | CD74         | in-frame     | five_prime   | 117646326          | 33              | 1853                        | True                   | lost          | 0.0                      | False               | Fibronectin type III domain                  | Protein tyrosine and serine/threonine kinase |                   | NA   Protein fusion: in frame (ROS1-CD74)                                                                                                                                                  | NA                            |
| EVT-P-0004336-T01-IM5-64 | P-0004336-T01-IM5 |            | ROS1-CD74   | CD74         | in-frame     | threep_prime | 117647249          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 with ROS1 exons 34-43) : t(5;6)(5q32;q22.1)(chr5:g.149783071::chr6:g.117647249)   Protein fusion: in frame (CD74-ROS1)       | NA                            |

| event_id                     | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                                 | disrupted_domains | source_annotation_text                                                                                                                                                                        | source_site2_effect_on_frame |
|------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|----------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>6213-T01-IM6-65 | P-0016<br>213-T01-IM6 |            | ROS1-CD74   | CD74         | out-of-frame | three_prime | 117649786          | 33              | 1789                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 33-43): t(5;6)(q32;q22.1)(chr5:g.149783418::chr6:g.117649786)   Protein Fusion: out of frame {CD74:ROS1}    | NA                           |
| EVT-P-000<br>2707-T01-IM3-66 | P-0002<br>707-T01-IM3 |            | ROS1-CD74   | CD74         | in-frame     | five_prime  | 117646946          | 33              | 1853                        | True                   | lost          | 0.0                      | False               | Fibronectin type III domain                  | Protein tyrosine and serine/threonine kinase |                   | NA   Protein fusion: in frame (ROS1-CD74)                                                                                                                                                     | NA                           |
| EVT-P-000<br>4747-T01-IM5-67 | P-0004<br>747-T01-IM5 |            | ROS1-CD74   | CD74         | out-of-frame | three_prime | 117649107          | 33              | 1789                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 33-43): t(5;6)(q32;q22.1)(chr5:g.149782561::chr6:g.117649107)   Protein fusion: out of frame (CD74-ROS1)    | NA                           |
| EVT-P-000<br>4790-T01-IM5-68 | P-0004<br>790-T01-IM5 |            | ROS1-CD74   | CD74         | in-frame     | three_prime | 117646467          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (exons 1-7 of CD74 fused to exons 34-43 of ROS1) : t(5;6)(q32;q22.1)(chr5:g.149782207::chr6:g.117646467)   Protein fusion: in frame (CD74-ROS1) | NA                           |
| EVT-P-003<br>6970-T01-IM6-69 | P-0036<br>970-T01-IM6 |            | ROS1-CD74   | CD74         | out-of-frame | three_prime | 117650273          | 33              | 1789                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q33.1;q22.1)(chr5:g.149783480::chr6:g.117650273)   Protein Fusion: out of frame {CD74:ROS1}                                             | NA                           |
| EVT-P-000<br>0632-T01-IM3-70 | P-0000<br>632-T01-IM3 |            | ROS1-CD74   | CD74         | in-frame     | three_prime | 117646313          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | NA   Protein fusion: in frame (CD74-ROS1)                                                                                                                                                     | NA                           |



| event_id                         | sample_id                 | patient_id | fusion_name       | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                                 | disrupted_domains | source_annotation_text                                                                                                                                                                     | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|-------------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|----------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>2722-T01-I<br>M6-71 | P-0032<br>722-T0<br>1-IM6 |            | ROS1<br>-CD7<br>4 | CD7<br>4     | unknown      | threonine_prime | 117645<br>677      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149782874:chr6:g.117645677)   Protein Fusion: mid-exon {CD74:ROS1}      | NA                           |
| EVT-P-001<br>4506-T01-I<br>M6-72 | P-0014<br>506-T0<br>1-IM6 |            | ROS1<br>-CD7<br>4 | CD7<br>4     | unknown      | threonine_prime | 117650<br>623      | 32              | 1750                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 32-43): t(5;6)(q32;q22.1)(chr5:g.149782871:chr6:g.117650623)   Protein fusion: mid-exon (CD74-ROS1)      | NA                           |
| EVT-P-001<br>8222-T01-I<br>M6-73 | P-0018<br>222-T0<br>1-IM6 |            | ROS1<br>-CD7<br>4 | CD7<br>4     | in-frame     | threonine_prime | 117646<br>209      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149783915:chr6:g.117646209)   Protein Fusion: in frame {CD74:ROS1}      | NA                           |
| EVT-P-000<br>1437-T01-I<br>M3-74 | P-0001<br>437-T0<br>1-IM3 |            | ROS1<br>-CD7<br>4 | CD7<br>4     | out-of-frame | threonine_prime | 117649<br>412      | 33              | 1789                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | None   Protein fusion: out of frame (CD74-ROS1)                                                                                                                                            | NA                           |
| EVT-P-001<br>8513-T01-I<br>M6-75 | P-0018<br>513-T0<br>1-IM6 |            | ROS1<br>-CD7<br>4 | CD7<br>4     | unknown      | threonine_prime | 117647<br>544      | 33              | 1800                        | False                  | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused with ROS1 exons 33 to 43): t(5;6)(q32;q22.1)(chr5:g.149783804:chr6:g.117647544)   Protein Fusion: mid-exon {CD74:ROS1} | NA                           |
| EVT-P-000<br>0569-T01-I<br>M3-76 | P-0000<br>569-T0<br>1-IM3 |            | ROS1<br>-CD7<br>4 | CD7<br>4     | out-of-frame | five_prime      | 117647<br>622      | 32              | 1789                        | True                   | lost          | 0.0                      | False               | Fibronectin type III domain                  | Protein tyrosine and serine/threonine kinase |                   | NA   Protein fusion: out of frame (ROS1-CD74)                                                                                                                                              | NA                           |

| event_id                         | sample_id                 | patient_id | fusion_name       | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                | disrupted_domains | source_annotation_text                                                                                                                                                                                 | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|------------|-------------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|-----------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-002<br>5107-T01-I<br>M6-77 | P-0025<br>107-T0<br>1-IM6 |            | ROS1<br>-CD7<br>4 | CD7<br>4     | in-frame     | threonine_prime | 117647<br>170      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149782204:chr6:g.117647170)   Protein Fusion: in frame {CD74:ROS1}                  | NA                            |
| EVT-P-000<br>0840-T01-I<br>M3-78 | P-0000<br>840-T0<br>1-IM3 |            | ROS1<br>-CD7<br>4 | CD7<br>4     | in-frame     | threonine_prime | 117646<br>938      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | ROS1 (NM_002944) - CD74 (NM_001025158) Translocation: t(6;5)(q22.1;q32)(chr6:g.117646938:chr5:g.149783508)   Protein fusion: in frame (CD74-ROS1)                                                      | NA                            |
| EVT-P-000<br>7108-T01-I<br>M5-79 | P-0007<br>108-T0<br>1-IM5 |            | ROS1<br>-CD7<br>4 | CD7<br>4     | in-frame     | threonine_prime | 117646<br>132      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1 to 6 fused with ROS1 exons 34 to 43) : t(5;6)(chr5q32;chr6q22.1)(chr5:g.149783704:chr6:g.117646132)   Protein fusion: in frame (CD74-ROS1) | NA                            |
| EVT-P-002<br>7196-T01-I<br>M6-80 | P-0027<br>196-T0<br>1-IM6 |            | ROS1<br>-CD7<br>4 | CD7<br>4     | out-of-frame | threonine_prime | 117650<br>377      | 33              | 1789                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 33-43): t(5;6)(q32;q22.1)(chr5:g.149782321:chr6:g.117650377)   Protein Fusion: out of frame {CD74:ROS1}              | NA                            |
| EVT-P-002<br>6589-T02-I<br>M6-81 | P-0026<br>589-T0<br>2-IM6 |            | ROS1<br>-CD7<br>4 | CD7<br>4     | in-frame     | threonine_prime | 117646<br>032      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) Fusion (CD74 exon6 fused to ROS1 exon34) : t(5;6)(q32;q22.1)(chr5:g.149782990:chr6:g.117646032)   Protein Fusion: in frame (CD74:ROS1)                          | NA                            |
| EVT-P-002<br>1054-T01-I<br>M6-82 | P-0021<br>054-T0<br>1-IM6 |            | ROS1<br>-CD7<br>4 | CD7<br>4     | in-frame     | threonine_prime | 117647<br>162      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149783021:chr6:g.117647162)   Protein Fusion: in frame {CD74:ROS1}                  | NA                            |

| event_id                         | sample_id                 | patient_id | fusion_name          | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                                 | disrupted_domains | source_annotation_text                                                                                                                                                                   | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|----------------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|----------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-000<br>8134-T01-I<br>M5-83 | P-0008<br>134-T0<br>1-IM5 |            | ROS1<br>-CD7<br>4    | CD7<br>4     | in-frame     | threonine_prime | 117646<br>080      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD274 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(5q32;q22.1)(chr5:g.149783121::chr6:g.117646080)   Protein fusion: in frame (CD74-ROS1) | NA                           |
| EVT-P-002<br>3617-T01-I<br>M6-84 | P-0023<br>617-T0<br>1-IM6 |            | ROS1<br>-CD7<br>4    | CD7<br>4     | in-frame     | threonine_prime | 117646<br>444      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | CD74 (NM_00102515) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149782946::chr6:g.117646444)   Protein Fusion: in frame {CD74:ROS1}    | NA                           |
| EVT-P-006<br>5104-T01-I<br>M7-86 | P-0065<br>104-T0<br>1-IM7 |            | ROS1<br>-CD7<br>4    | CD7<br>4     | in-frame     | five_prime      | 117646<br>386      | 33              | 1853                        | True                   | lost          | 0.0                      | False               | Fibronectin type III domain                  | Protein tyrosine and serine/threonine kinase |                   | ROS1 (NM_002944) - CD74 (NM_001025159) rearrangement: t(5;6)(q32;q22.1)(chr5:g.149783694::chr6:g.117646386)   Protein Fusion: in frame {ROS1:CD74}                                       | NA                           |
| EVT-P-004<br>1118-T07-I<br>M7-87 | P-0041<br>118-T0<br>7-IM7 |            | ROS1<br>-CD7<br>4    | CD7<br>4     | out-of-frame | five_prime      | 117648<br>241      | 32              | 1789                        | True                   | lost          | 0.0                      | False               | Fibronectin type III domain                  | Protein tyrosine and serine/threonine kinase |                   | ROS1 (NM_002944) - CD74 (NM_001025159) rearrangement: t(5;6)(q32;q22.1)(chr5:g.149782213::chr6:g.117648241)   Protein Fusion: out of frame {ROS1:CD74}                                   | NA                           |
| EVT-P-000<br>8996-T04-I<br>M6-88 | P-0008<br>996-T0<br>4-IM6 |            | ROS1<br>-CEP<br>85L  | CEP<br>85L   | unknown      | threonine_prime | 117687<br>428      | 18              | 875                         | False                  | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | CEP85L (NM_001178035) - ROS1 (NM_002944) rearrangement : c.242-19954:CEP85L_c.2623:ROS1del   Protein Fusion: mid-exon {CEP85L:ROS1}                                                      | NA                           |
| EVT-P-005<br>5135-T03-I<br>M7-90 | P-0055<br>135-T0<br>3-IM7 |            | ROS1<br>-COL<br>13A1 | COL<br>13A1  | in-frame     | five_prime      | 117658<br>665      | 30              | 1693                        | True                   | lost          | 0.0                      | False               | Fibronectin type III domain                  | Protein tyrosine and serine/threonine kinase |                   | ROS1 (NM_002944) - COL13A1 (NM_001130103) rearrangement: t(6;10)(q22.1;q22.1)(chr6:g.117658665::chr10:g.71715849)   Protein Fusion: in frame {ROS1:COL13A1}                              | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                | disrupted_domains | source_annotation_text                                                                                                                                            | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|-----------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-002<br>0458-T01-I<br>M6-98  | P-0020<br>458-T0<br>1-IM6 |            | ROS1<br>-EZR | EZR          | in-frame     | three_prime | 117646<br>033      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | EZR (NM_003379) - ROS1 (NM_002944) fusion (EZR exons 1-9 fused with ROS1 exons 34-43): c.1090+207:ER_c.5558-455:ROS1del   Protein Fusion: in frame {EZR:ROS1}     | NA                            |
| EVT-P-000<br>6047-T02-I<br>M5-99  | P-0006<br>047-T0<br>2-IM5 |            | ROS1<br>-EZR | EZR          | in-frame     | three_prime | 117647<br>241      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | EZR (NM_003379) - ROS1 (NM_002944) fusion (EZR exons 1-9 with ROS1 exons 34-43): c.1091-355:EZR_c.5557+146:ROS1del   Protein fusion: in frame (EZR-ROS1)          | NA                            |
| EVT-P-000<br>1117-T01-I<br>M3-100 | P-0001<br>117-T0<br>1-IM3 |            | ROS1<br>-EZR | EZR          | unknown      | three_prime | 117650<br>595      | 32              | 1755                        | False                  | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | EZR (NM_003379) - ROS1 (NM_002944) Deletion: c.1090+214_c.5263del   Protein fusion: mid-exon (EZR-ROS1)                                                           | NA                            |
| EVT-P-001<br>0989-T01-I<br>M5-101 | P-0010<br>989-T0<br>1-IM5 |            | ROS1<br>-EZR | EZR          | out-of-frame | three_prime | 117649<br>729      | 33              | 1789                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | EZR (NM_003379) - ROS1 (NM_002944) fusion (EZR exons 1-9 fused with ROS1 exons 33-43): c.1090+215:ER_c.5366+763:ROS1del   Protein fusion: out of frame (EZR-ROS1) | NA                            |
| EVT-P-000<br>6047-T03-I<br>M6-102 | P-0006<br>047-T0<br>3-IM6 |            | ROS1<br>-EZR | EZR          | in-frame     | three_prime | 117647<br>241      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | EZR (NM_003379) - ROS1 (NM_002944) fusion: c.1091-356:EZR_c.5557+146:ROS1del   Protein Fusion: in frame {EZR:ROS1}                                                | NA                            |
| EVT-P-000<br>3970-T01-I<br>M5-103 | P-0003<br>970-T0<br>1-IM5 |            | ROS1<br>-EZR | EZR          | out-of-frame | three_prime | 117649<br>698      | 33              | 1789                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | ROS1 (NM_002944) - EZR (NM_003379) rearrangement: c.1090+142 C:EZR_c.5366+794T:ROS1del   Protein fusion: out of frame (EZR-ROS1)                                  | NA                            |
| EVT-P-001<br>4437-T01-I<br>M6-104 | P-0014<br>437-T0<br>1-IM6 |            | ROS1<br>-EZR | EZR          | out-of-frame | three_prime | 117650<br>363      | 33              | 1789                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | ROS1 (NM_002944) _EZR (NM_003379) rearrangement: c.1090+229:ER_c.5366+129:ROS1del   Protein fusion: out of frame (EZR-ROS1)                                       | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                                 | disrupted_domains | source_annotation_text                                                                                                                                                | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|----------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-003<br>4609-T01-I<br>M6-105 | P-0034<br>609-T0<br>1-IM6 |            | ROS1<br>-EZR    | EZR          | in-frame     | threonine_prime | 117647<br>244      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | EZR (NM_003379) - ROS1 (NM_002944) fusion: c.1090+175:EZR_c.5557+143:ROS1del   Protein Fusion: in frame {EZR:ROS1}                                                    | NA                            |
| EVT-P-001<br>4304-T02-I<br>M6-106 | P-0014<br>304-T0<br>2-IM6 |            | ROS1<br>-EZR    | EZR          | in-frame     | threonine_prime | 117645<br>949      | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | EZR (NM_003379) - ROS1 (NM_002944) fusion: (EZR exons 1 to 9 fused with ROS1 exons 34 to 43): c.1090+79:EZR_c.5558-371:ROS1del   Protein Fusion: in frame {EZR:ROS1}  | NA                            |
| EVT-P-000<br>4445-T01-I<br>M5-107 | P-0004<br>445-T0<br>1-IM5 |            | ROS1<br>-GOLGB1 | GOLGB1       | in-frame     | threonine_prime | 117645<br>253      | 35              | 1881                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | GOLGB1 (NM_004487) - ROS1 (NM_002944) rearrangement : t(3;6)(3q13.33;6q22.1)(chr3:121434446::chr6:117645253)   Protein fusion: in frame (GOLGB1-ROS1)                 | NA                            |
| EVT-P-000<br>5086-T01-I<br>M5-108 | P-0005<br>086-T0<br>1-IM5 |            | ROS1<br>-GOPC   | GOPC         | in-frame     | threonine_prime | 117644<br>378      | 35              | 1881                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | GOPC(NM_020399) -ROS1(NM_002944) - Fusion (GOPC exon 8 fused to ROS1 exon 34) : c.1259-1533:GOPC_c.5641+1117:ROS1del   Protein fusion: in frame (GOPC-ROS1)           | NA                            |
| EVT-P-000<br>7626-T01-I<br>M5-109 | P-0007<br>626-T0<br>1-IM5 |            | ROS1<br>-GOPC   | GOPC         | in-frame     | threonine_prime | 117644<br>860      | 35              | 1881                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                  |                   | GOPC (NM_020399) - ROS1 (NM_002944) fusion (GOPC exons 1-8 fused with ROS1 exons 35-43) : c.1259-1217: GOPC_c.5641+635:ROS1del   Protein fusion: in frame (GOPC-ROS1) | NA                            |
| EVT-P-001<br>9111-T01-I<br>M6-111 | P-0019<br>111-T0<br>1-IM6 |            | ROS1<br>-GOPC   | GOPC         | out-of-frame | five_prime      | 117643<br>452      | 34              | 1881                        | True                   | lost          | 0.0                      | False               | Fibronectin type III domain                  | Protein tyrosine and serine/threonine kinase |                   | ROS1 (NM_002944) - GOPC (NM_020399) rearrangement: c.5642-895: ROS1_c.912+138:GOPCdup   Protein Fusion: out of frame {ROS1:GOPC}                                      | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name      | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                                          | lost_domains                                 | disrupted_domains | source_annotation_text                                                                                                                                  | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------------------------------------|----------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-005<br>8302-T02-I<br>M7-112 | P-0058<br>302-T0<br>2-IM7 |            | ROS1<br>-GRIK2   | GRIK2        | unknown      | five_prime  | 117622188          | 42              | 2228                        | False                  | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase; Fibronectin type III domain |                                              |                   | ROS1 (NM_002944) - GRIK2 (NM_021956) fusion: c.6682:ROS1_c.2086-17008:GRIK2inv   Protein Fusion: mid-exon {ROS1:GRIK2}                                  | NA                            |
| EVT-P-002<br>1914-T01-I<br>M6-113 | P-0021<br>914-T0<br>1-IM6 |            | ROS1<br>-LARS1   | LARS1        | out-of-frame | five_prime  | 117646519          | 33              | 1853                        | True                   | lost          | 0.0                      | False               | Fibronectin type III domain                                               | Protein tyrosine and serine/threonine kinase |                   | ROS1 (NM_002944) - LARS (NM_020117) rearrangement: t(5;6)(q32,q22.1)(chr5:g.145500267::chr6:g.117646519)   Protein Fusion: out of frame {ROS1:LARS}     | NA                            |
| EVT-P-003<br>6702-T01-I<br>M6-118 | P-0036<br>702-T0<br>1-IM6 |            | ROS1<br>-NUS1    | NUS1         | unknown      | three_prime | 117711069          | 12              | 421                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase                              | Fibronectin type III domain                  |                   | ROS1 (NM_002944) rearrangement: c.1263-60:ROS1_chr6:g.118032766inv   Transcript Fusion {NUS1:ROS1}                                                      | NA                            |
| EVT-P-001<br>6046-T01-I<br>M6-171 | P-0016<br>046-T0<br>1-IM6 |            | ROS1<br>-SLC34A2 | SLC34A2      | out-of-frame | three_prime | 117662059          | 31              | 1694                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase                              | Fibronectin type III domain                  |                   | ROS1 (NM_002944) - SLC34A2(NM_006424) fusion: t(4;6) (p15;q22)(chr4:g.25667671::chr6:g.117662059)   Protein Fusion: out of frame {SLC34A2:ROS1}         | NA                            |
| EVT-P-001<br>0984-T01-I<br>M5-175 | P-0010<br>984-T0<br>1-IM5 |            | ROS1<br>-SLC4A4  | SLC4A4       | in-frame     | three_prime | 117646719          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase                              | Fibronectin type III domain                  |                   | SLC4A4 (NM_001134742) - ROS1 (NM_002944) rearrangement: t(4;6)(q13.3;q22.1)(chr4:g.72429843::chr6:g.117646719)   Protein fusion: in frame (SLC4A4-ROS1) | NA                            |
| EVT-P-001<br>3451-T01-I<br>M5-178 | P-0013<br>451-T0<br>1-IM5 |            | ROS1<br>-STX7    | STX7         | out-of-frame | three_prime | 117649456          | 33              | 1789                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase                              | Fibronectin type III domain                  |                   | STX7 (NM_003569) - ROS1 (NM_002944) rearrangement: c.86-3411:STX7_c.5366+1036:ROS1del   Protein fusion: out of frame (STX7-ROS1)                        | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name   | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                | disrupted_domains | source_annotation_text                                                                                                                                                               | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|---------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|-----------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>9670-T03-I<br>M6-180 | P-0019<br>670-T0<br>3-IM6 |            | ROS1<br>-TFG  | TFG          | in-frame     | threonine_prime | 117643761          | 35              | 1881                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | TFG (NM_001195478) - ROS1 (NM_002944) fusion (TFG exon 4 fused to ROS1 exon 35): t(3;6)(q12.2;q22.1)(chr3:g.100448554::chr6:g.17643761)   Protein Fusion: in frame {TFG:ROS1}        | NA                           |
| EVT-P-001<br>9670-T02-I<br>M6-181 | P-0019<br>670-T0<br>2-IM6 |            | ROS1<br>-TFG  | TFG          | in-frame     | threonine_prime | 117643761          | 35              | 1881                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | TFG (NM_001195478) - ROS1 (NM_002944) fusion (TFG exon 4 fused to ROS1 exon 35): t(3;6)(q12.2;q22.1)(chr3:g.100448554::chr6:g.17643761)   Protein Fusion: in frame {TFG:ROS1}        | NA                           |
| EVT-P-000<br>3207-T01-I<br>M5-183 | P-0003<br>207-T0<br>1-IM5 |            | ROS1<br>-TPM3 | TPM3         | in-frame     | threonine_prime | 117642993          | 35              | 1881                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | NA   Protein fusion: in frame (TPM3-ROS1)                                                                                                                                            | NA                           |
| EVT-P-002<br>8865-T02-I<br>M6-184 | P-0028<br>865-T0<br>2-IM6 |            | ROS1<br>-TPM3 | TPM3         | in-frame     | threonine_prime | 117643589          | 35              | 1881                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | TPM3 (NM_001278190) - ROS1 (NM_002944) fusion (TPM3 exons 1-6 with ROS1 exons 35-43): t(1;6)(q21.3;q22.1)(chr1:g.154133648::chr6:g.117643589)   Protein Fusion: in frame {TPM3:ROS1} | NA                           |
| EVT-P-000<br>1840-T01-I<br>M3-185 | P-0001<br>840-T0<br>1-IM3 |            | SDC4<br>-ROS1 | SDC4         | in-frame     | threonine_prime | 117650688          | 32              | 1750                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | SDC4 (NM_002999) - ROS1 (NM_002944) Translocation: t(20;6)(q13.12;q22.1)(chr20:43962892::chr6:117650688)   Protein fusion: in frame (SDC4-ROS1)                                      | NA                           |
| EVT-P-000<br>1840-T02-I<br>M5-186 | P-0001<br>840-T0<br>2-IM5 |            | SDC4<br>-ROS1 | SDC4         | in-frame     | threonine_prime | 117650732          | 32              | 1750                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(20;6)(q13.12;q22.1)(chr20:g.43962893::chr6:g.117650732)   Protein fusion: in frame (SDC4-ROS1)                                         | NA                           |
| EVT-P-005<br>4419-T02-I<br>M6-187 | P-0054<br>419-T0<br>2-IM6 |            | SDC4<br>-ROS1 | SDC4         | unknown      | threonine_prime | 117645734          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117645734::chr20:g.43955928)   Protein Fusion: mid-exon {SDC4:ROS1}                                         | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name | partner_gene | frame_status | target_role  | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                | disrupted_domains | source_annotation_text                                                                                                                                                                    | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-------------|--------------|--------------|--------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|-----------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-005<br>4419-T01-I<br>M6-188 | P-0054<br>419-T0<br>1-IM6 |            | SDC4-ROS1   | SDC4         | unknown      | threep_prime | 117645734          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117645734::chr20:g.43955928)   Protein Fusion: mid-exon {SDC4:ROS1}                                              | NA                            |
| EVT-P-004<br>7840-T02-I<br>M6-189 | P-0047<br>840-T0<br>2-IM6 |            | SDC4-ROS1   | SDC4         | unknown      | threep_prime | 117645871          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117645871::chr20:g.43955947)   Protein Fusion: mid-exon {SDC4:ROS1}                                              | NA                            |
| EVT-P-000<br>4414-T01-I<br>M5-190 | P-0004<br>414-T0<br>1-IM5 |            | SDC4-ROS1   | SDC4         | in-frame     | threep_prime | 117651952          | 32              | 1750                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-4 fused with ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117651952::chr20:g.43957039)   Protein fusion: in frame (SDC4-ROS1) | NA                            |
| EVT-P-004<br>3938-T01-I<br>M6-191 | P-0043<br>938-T0<br>1-IM6 |            | SDC4-ROS1   | SDC4         | in-frame     | threep_prime | 117651208          | 32              | 1750                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117651208::chr20:g.43962372)   Protein Fusion: in frame {SDC4:ROS1}                                              | NA                            |
| EVT-P-000<br>4946-T02-I<br>M5-192 | P-0004<br>946-T0<br>2-IM5 |            | SDC4-ROS1   | SDC4         | in-frame     | threep_prime | 117651013          | 32              | 1750                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117651013::chr20:g.43962011)   Protein fusion: in frame (SDC4-ROS1)                                              | NA                            |
| EVT-P-003<br>8394-T01-I<br>M6-193 | P-0038<br>394-T0<br>1-IM6 |            | SDC4-ROS1   | SDC4         | in-frame     | threep_prime | 117654360          | 32              | 1750                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | SDC4 (NM_002999) - ROS1 (NM_002944) rearrangement: t(6;20)(q22.1;q13.12)(chr6:g.117654360::chr20:g.43961742)   Protein Fusion: in frame {SDC4:ROS1}                                       | NA                            |
| EVT-P-002<br>7630-T01-I<br>M6-194 | P-0027<br>630-T0<br>1-IM6 |            | SDC4-ROS1   | SDC4         | in-frame     | threep_prime | 117651891          | 32              | 1750                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-2 fused to ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117651891::chr20:g.43962429)   Protein Fusion: in frame {SDC4:ROS1}   | NA                            |

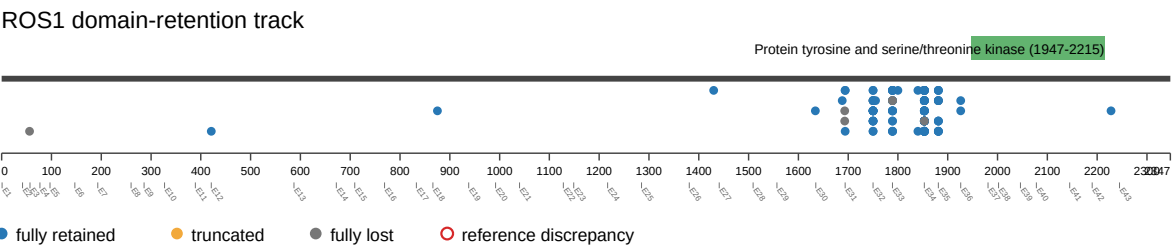


| event_id                          | sample_id                 | patient_id | fusion_name | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                | disrupted_domains | source_annotation_text                                                                                                                                                                           | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|-----------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-006<br>4541-T01-I<br>M7-195 | P-0064<br>541-T0<br>1-IM7 |            | SDC4-ROS1   | SDC4         | in-frame     | threonine_prime | 117650669          | 32              | 1750                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117650669::chr20:g.43962537)   Protein Fusion: in frame {SDC4:ROS1}                                                     | NA                            |
| EVT-P-002<br>2578-T02-I<br>M6-196 | P-0022<br>578-T0<br>2-IM6 |            | SDC4-ROS1   | SDC4         | in-frame     | threonine_prime | 117652158          | 32              | 1750                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-2 fused in-frame to ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117652158::chr20:g.43962266)   Protein Fusion: in frame {SDC4:ROS1} | NA                            |
| EVT-P-002<br>2578-T01-I<br>M6-197 | P-0022<br>578-T0<br>1-IM6 |            | SDC4-ROS1   | SDC4         | in-frame     | threonine_prime | 117652158          | 32              | 1750                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-2 fused in-frame to ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117652158::chr20:g.43962266)   Protein Fusion: in frame {SDC4:ROS1} | NA                            |
| EVT-P-001<br>2008-T01-I<br>M5-198 | P-0012<br>008-T0<br>1-IM5 |            | SDC4-ROS1   | SDC4         | in-frame     | threonine_prime | 117652119          | 32              | 1750                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | SDC4 (NM_002999) - ROS1 (NM_002944) reciprocal fusion (SDC4 exons 1-2 with ROS1 exons 32-43): t(6;20)(q22.1;q13.12) (chr6:g.117652119::chr20:g.43964016 )   Protein fusion: in frame (SDC4-ROS1) | NA                            |
| EVT-P-001<br>7325-T01-I<br>M6-199 | P-0017<br>325-T0<br>1-IM6 |            | SDC4-ROS1   | SDC4         | in-frame     | threonine_prime | 117654375          | 32              | 1750                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-2 fused in-frame to ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117654375::chr20:g.43963860)   Protein Fusion: in frame {SDC4:ROS1} | NA                            |
| EVT-P-001<br>2008-T02-I<br>M6-200 | P-0012<br>008-T0<br>2-IM6 |            | SDC4-ROS1   | SDC4         | in-frame     | threonine_prime | 117652099          | 32              | 1750                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain |                   | SDC4 (NM_002999) - ROS1 (NM_002944) reciprocal fusion (SDC4 exons 1-2 with ROS1 exons 32-43): t(6;20)(q22.1;q13.12) (chr6:g.117652119::chr20:g.43964016)   Protein Fusion: in frame {SDC4:ROS1}  | NA                            |

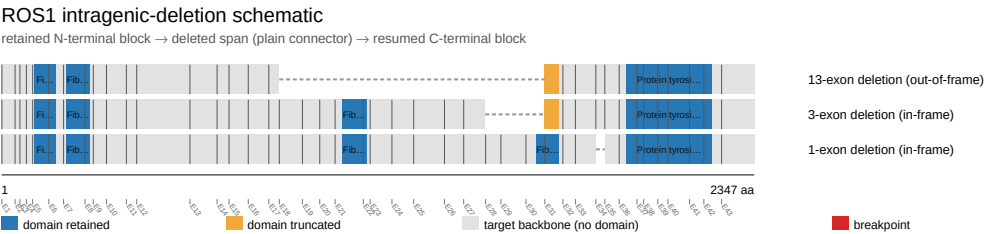
| event_id                          | sample_id                 | patient_id | fusion_name   | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                                                              | disrupted_domains | source_annotation_text                                                                                                                                                                           | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|---------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|---------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>6504-T01-I<br>M6-201 | P-0016<br>504-T0<br>1-IM6 |            | SDC4-ROS1     | SDC4         | in-frame     | threep_rime | 117650681          | 32              | 1750                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                                               |                   | SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-2 fused in-frame to ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117650681::chr20:g.43961904)   Protein Fusion: in frame {SDC4:ROS1} | NA                           |
| EVT-P-000<br>3507-T01-I<br>M5-202 | P-0003<br>507-T0<br>1-IM5 |            | SLC16A10-ROS1 | SLC16A10     | in-frame     | five_p_rime | 117739561          | 2               | 56                          | True                   | lost          | 0.0                      | False               |                                              | Protein tyrosine and serine/threonine kinase; Fibronectin type III domain |                   | NA   Protein fusion: in frame (ROS1-SLC16A10)                                                                                                                                                    | NA                           |
| EVT-P-004<br>2481-T01-I<br>M6-203 | P-0042<br>481-T0<br>1-IM6 |            | SLC4A4-ROS1   | SLC4A4       | out-of-frame | threep_rime | 117646662          | 34              | 1853                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                                               |                   | SLC4A4 (NM_001134742) - ROS1 (NM_002944) fusion: t(4;6)(q13.3;q22.1)(chr4:g.72430718::chr6:g.117646662)   Protein Fusion: out of frame {SLC4A4:ROS1}                                             | NA                           |
| EVT-P-004<br>5322-T02-I<br>M6-205 | P-0045<br>322-T0<br>2-IM6 |            | TFG-ROS1      | TFG          | in-frame     | threep_rime | 117644683          | 35              | 1881                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase | Fibronectin type III domain                                               |                   | TFG (NM_001195478) - ROS1 (NM_002944) fusion: t(3;6)(q12.2;q22.1)(chr3:g.100448076::chr6:g.117644683)   Protein Fusion: in frame {TFG:ROS1}                                                      | NA                           |
| EVT-P-000<br>7870-T02-I<br>M6-207 | P-0007<br>870-T0<br>2-IM6 |            | ZNF157-ROS1   | ZNF157       | unknown      | five_p_rime | 117659405          | 30              | 1693                        | True                   | lost          | 0.0                      | False               | Fibronectin type III domain                  | Protein tyrosine and serine/threonine kinase                              |                   | ROS1 (NM_002944) Rearrangement : t(X;6)(p11.23;q22.1)(chrX:g.47262831; chr6:g.117659405)   Antisense Fusion                                                                                      | NA                           |

# Figures

## domain retention outliers



# intragenic deletion schematic



# reference comparison

Reference vs msk\_impact\_50k\_2026

